

JARED HWANG

Curriculum Vitae: Updated April 2026

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Education

- 09/2024 - Present **University of Washington**, Seattle, WA
Ph.D. Computer Science & Engineering
Advisor: Prof. Jon E. Froehlich
- 06/2021 - 08/2023 **University of Southern California**, Los Angeles, CA
M.S. Computer Science
GPA: 4.00, M.S. Honors Program
- 08/2016 - 05/2020 **Tufts University**, Medford, MA
B.S. Physics, B.S. Computer Science
GPA: 3.94, summa cum laude, Sigma Pi Sigma Honor Society

Conference Publications

- [C3] **BikeButler: A Personalized, Context-sensitive Bike Routing Tool using Open Data and VLM-based Analyses of Street View Imagery** [pdf]
Jared Hwang, John O'Meara, Zeyu Wang, Jasmine Zhang, Jon E. Froehlich.
In Proceedings of CHI'26
- [C2] **"Where can I Park?" Understanding Human Perspectives and Scalably Detecting Disability Parking from Aerial Imagery** [pdf]
Jared Hwang, Chu Li, Hanbyul Kang, Maryam Hosseini, Jon E. Froehlich.
In Proceedings of ASSETS'25
- [C1] **Digital Typhoon: Long-term Satellite Image Dataset for the Spatio-Temporal Modeling of Tropical Cyclones** [pdf] [Web] [Code]
Asanobu Kitamoto, Jared Hwang, Bastien Vuillod, Lucas Gautier, Yingtao Tian, Tarin Clanuwat.
In Proceedings of NeurIPS'23

Workshops, Posters, and Extended Abstracts

- [W2] **RAMPNET: A Two-Stage Pipeline for Bootstrapping Curb Ramp Detection in Streetscape Images from Open Government Metadata** [pdf]
John S. O'Meara, Jared Hwang, Zeyu Wang, Michael Saugstad, Jon E. Froehlich.
ICCV'25 Workshop on Vision Foundation Models and Generative AI for Accessibility

[W1]

“Does the cafe entrance look accessible? Where is the door?” Towards Geospatial AI Agents for Visual Inquiries [\[pdf\]](#)

Jon E. Froehlich, **Jared Hwang**, Zeyu Wang, John S. O'Meara, Xia Su, William Huang, Yang Zhang, Alex Fiannaca, Philip Nelson, Shaun Kane.

ICCV'25 Workshop on Vision Foundation Models and Generative AI for Accessibility

Research Experience

09/2024 - Present

UW Makeability Lab, University of Washington, Seattle, WA, USA

Advisor: Prof. Jon E. Froehlich

- Created *BikeButler*, a bike-routing platform that allows users to generate, view, customize, and save bicycle routes based on fine-grained preferences on eight 'bikeability' features. Conducted 16 user-study interviews. [\[C3\]](#)
- Developed and evaluated *AccessParkCV*, a CV pipeline that automatically detects and measures the width of disability parking spaces from aerial imagery. Created and released an 11,762 object dataset, and conducted 11 needfinding interviews [\[C2\]](#)

01/2023 - 08/2023

NII Kitamoto Lab, National Institute of Informatics (NII), Tokyo, Japan

Advisor: Prof. Asanobu Kitamoto

Designed and built Pyphoon2, a dataloader for the Digital Typhoon Dataset. Designed and ran ML benchmark experiments, including classification, regression, and forecasting of typhoon intensity. Work published to NeurIPS 2023 Datasets and Benchmarks. [\[C1\]](#)

05/2020 - 09/2021

Tufts Neutrino Group, Tufts University, Medford, MA, USA

Advisor: Prof. Taritree Wongjirad

Designed and ran ML experiments on neutrino detector image datasets. Created autoencoder based image compression model, producing 80% smaller filesizes compared to JPEG. Prototyped spatial loss algorithm for detector image segmentation.

06/2018 - 03/2020

MIT Laboratory for Nuclear Science, MIT, Cambridge, MA, USA

Advisor: Prof. Janet Conrad and Dr. Daniel Winklehner

Developed a particle accelerator simulation code for radio frequency quadrupoles (RFQ) in the IsoDAR experiment. Ran simulations with greater accuracy and 20% faster than competing solutions. Included visualization and parallel computing capability. [\[Code\]](#)

09/2017 - 05/2018

Tufts Sajina Lab, Tufts University, Medford, MA

Advisor: Prof. Anna Sajina

Performed statistical analysis and crossmatching of two astronomical satellite datasets.

Projects

11/2021 - 12/2021

Simulation Inference on Urban Traffic Data

University of Southern California, Los Angeles, CA

Applied Bayesian conditional density estimation simulation inference to traffic simulations of the Seattle area. [\[Web\]](#)

Industry Experience

- 10/2023 - 10/2024 **Software Development Engineer**
Amazon, Seattle, WA
Maintained massive pipelines and databases for customer behavior and purchase data.
- 05/2022 - 08/2022 **Software Development Engineer Internship**
Amazon, Seattle, WA
Integrated Apache Airflow workflow orchestrator for ML pipelines.

Invited Talks

- 10/2025 **Pacific Northwest Transportation Consortium (PacTrans) '25**
BikeButler: Creating and Previewing Personalized, Context-Sensitive Bicycle Routes
- 06/2023 **Fujitsu & YNU Typhoon Research Lab Joint Research Workshop**
Machine Learning for the Digital Typhoon Dataset

Service

Community

- UW HCI Visit Days Scheduler ('26)

Paper Reviews

- DIS'26, ICCV'25 Workshop on Vision Models and Generative AI for Accessibility

Teaching Assistant

- UW CSE160 Data Programming (Wi'25, Fa'25)
- USC CSCI576 Multimedia Systems Design (Fa'22)
- Tufts COMP105 Programming Languages (Sp'20)
- Tufts COMP40: Machine Structure and Assembly Language Programming (Fa'18, Sp'20)

Honors

- 05/2020 **N. Hobbs Knight Prize Scholarship**
Tufts University, Medford, MA
For demonstrating outstanding ability in theoretical and experimental physics

Technical Skills

Programming

Python, C++, C, Java, JavaScript, SQL, R

Machine Learning & Development

PyTorch, Tensorflow, PyTorch Lightning, Docker, Slurm, Apptainer

Language

Conversational Japanese, Conversational Mandarin